Completed by fourth-year student

"Computing and Digital Technologies"

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Laboratory Work No. 2

Section A (Implementation and Results)

The practical part of this laboratory work was completed in two parts. First, I implemented a Flask REST API. GET /time returned the current server time as JSON with status 200, while POST /echo returned the message sent by the client. If the message field was missing, the server returned 400 Bad Request. Each /echo request was handled independently, so the server did not store previous messages. Because port 5000 was already in use on macOS, the API was run on port 5001.

```
(.venv) sofiyaserbina@MacBook-Pro-Sofiya Lab2_REST_Queues % curl http://127.0.0.1:5001/time
{"server_time":"2026-10-09 15:47:55.598426"}

(.venv) sofiyaserbina@MacBook-Pro-Sofiya Lab2_REST_Queues % curl -X POST http://127.0.0.1:5001/echo \
-H "Content-Type: application/json" \
-d '{"message": "magic!"}'
{"message":"magic!"}

(.venv) sofiyaserbina@MacBook-Pro-Sofiya Lab2_REST_Queues % curl -i -X POST http://127.0.0.1:5001/echo \
-H "Content-Type: application/json" \
-d '{}'
HTTP/1.1 400 BAD REQUEST
Server: Werkzeug/3.1.9 Python/3.9.6
Date: Fri, 09 Oct 2026 10:49:20 GMT
Content-Type: application/json
Content-Length: 40
Connection: close

{"error":"field 'message' is required"}
```

Figure 1. Testing GET /time, successful POST /echo, and the 400 error response.

In the second part, I implemented an in-memory message queue broker on port 6000. POST /queue/<topic> added a message and GET /queue/<topic> removed the oldest one, preserving FIFO order. The producer sent about one order per second, while the consumer simulated about three seconds of processing per order. This made the queue grow and demonstrated that the producer and consumer can work independently.

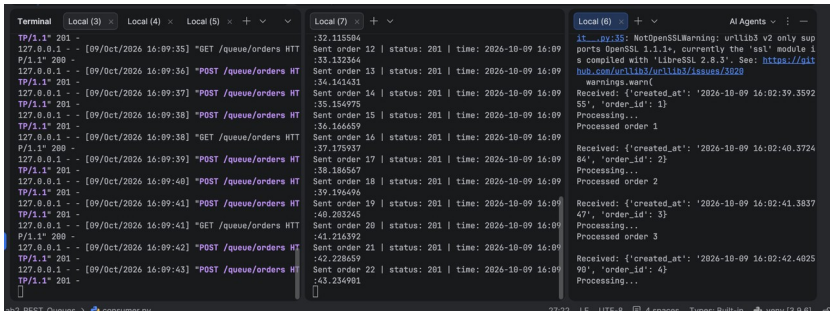


Figure 2. Producer and consumer running at different rates through the queue broker.

Reflection Questions

1. How does a REST request fundamentally differ from the TCP session in Lab 1 in terms of connection duration?

- In Lab 1, the client and server kept one TCP connection open while several commands were exchanged. In this lab, each REST operation is a separate HTTP request-response interaction. The server processes each request independently and does not need to remember earlier /echo requests.

2. Why does /echo use POST rather than GET? What would change if you used GET with parameters in the URL?

- POST allows the message to be sent in the JSON request body. With GET, the value would normally be placed in the URL, for example /echo?message=hello. That would expose the input in the query string and would be less suitable for submitting structured data.

3. What happens to the messages in the orders queue if you stop consumer.py for 30 seconds while producer.py keeps running?

- I stopped the consumer while leaving the producer and broker running. New orders continued to accumulate in the queue. After the consumer was started again, it processed the older messages in FIFO order. The messages were preserved because the broker process stayed active.

4. Give an example of a real situation where a synchronous REST call would be the better choice over a message queue, and vice versa.

- REST is suitable when an immediate response is required, such as checking an account balance. A message queue is better for background work, such as sending confirmation emails or processing orders, because the sender can continue without waiting for the consumer to finish.